# Sanchayan Dutta

*Curriculum Vitae (updated November 2026)*

*Department of Mathematics*
*University of California, Davis*
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## Education

2022–present **Ph.D. candidate**, *University of California, Davis*, Mathematics
Advised by Greg Kuperberg. Advanced to candidacy June 17, 2024.
Cumulative UC GPA 3.841/4.000.

2021–2022 **Graduate study**, *University of California, Davis*, Computer Science
Entered UC Davis in Fall 2021 and transferred to Mathematics in 2022.

2017–2021 **B.E.**, *Jadavpur University*, Electronics and Telecommunication Engineering

## Research interests

My research is in quantum coding theory, computational mathematics, and theoretical computer science. I study the classification and enumeration of classical and quantum codes using algebraic and computational methods. I also work on the mathematical foundations of machine learning and optimization on manifolds.

## Research experience

Summer 2024 **Research intern**, *Technical University of Munich*
Supervised by Prof. Suvrit Sra. Research on the mathematics of transformers and learning optimization algorithms.

Summer 2023 **Research intern**, *Harvard University*, School of Engineering and Applied Sciences
Supervised by Prof. Anurag Anshu. Research on sum-of-squares lower bounds and the no low-energy trivial states theorem.

Summer 2018 **Undergraduate research assistant**, *IISER Kolkata*, Department of Physical Sciences
Summer Student Research Program. Research on quantum algorithms and quantum circuit design.

## Writing

Publication titles and arXiv identifiers link to the corresponding papers.

9. **Enumerating Pauli Stabilizer Codes through Ten Qubits.** With Greg Kuperberg. Preprint (2026).
8. **Isometry Classes of Codes via Hall Polynomials.** Preprint (2026).
7. **Active Inference as Context Acquisition for AI Agents.** With Sai Niranjan Ramachandran and Suvrit Sra. Preprint (2026). arXiv 2608.19202.
6. **A Note on Clifford Stabilizer Codes for Ising Anyons.** Preprint (2025). arXiv 2503.08736.
5. **Toward Generalizable Learning of All (Linear) First-Order Methods via Memory Augmented Transformers.** With Suvrit Sra. Preprint (2024, revised 2025). arXiv 2410.07263.
4. **Riemannian Bilevel Optimization.** With Xiang Cheng and Suvrit Sra. Preprint (2024). Under review at *SIAM Journal on Mathematics of Data Science*. arXiv 2405.15816.
3. **Quantum Machine Learning: A Review and Current Status.** With N. Mishra et al. *Data Management, Analytics and Innovation*, Advances in Intelligent Systems and Computing **1175**, 101–145 (2021, first published online 2020).

2. **Quantum Circuit Design Methodology for Multiple Linear Regression.** With Adrien Suau, Sagnik Dutta, Suvadeep Roy, Bikash K. Behera, and Prasanta K. Panigrahi. *IET Quantum Communication* **1**(2), 55–61 (2020).
1. **Euler Number and Percolation Threshold on a Square Lattice With Diagonal Connection Probability and Revisiting the Island-Mainland Transition.** With Sugata Sen, Tajkera Khatun, Tapati Dutta, and Sujata Tarafdar. *Frontiers in Physics* **7**, Article 61 (2019).

## Talks

June 2023 **SoS Lower Bounds, SS-HDX and NLTS.** Anurag Anshu's research group, Harvard SEAS.

Nov. 2022 **An Introduction to Homological Codes with an Emphasis on Fourier and Poincaré Dualities.** Isaac Kim's research group, UC Davis.

## Service

2022 Co-organized the UC Davis Quantum Algebra & Computing Seminar Group with Daniel Tobias.

- Co-organized the UC Davis Computational Complexity & PCPs Reading Group with Daniel Tobias.

## Teaching

### University of California, Davis

**Teaching assistant**

- 2026 Fall
  - MAT 17B, Calculus for Biology and Medicine. With Dowman Varn.
- 2026 Winter
  - MAT 21B, Integral Calculus. With Dowman Varn.
- 2025 Fall
  - MAT 21B, Integral Calculus. With Dowman Varn.
- 2025 Winter
  - MAT 21B, Integral Calculus. With Jesús De Loera.
- 2024 Fall
  - MAT 21A, Differential Calculus. With John Challenor and Yoon Nho.
- 2024 Winter
  - MAT 21C, Partial Derivatives and Series. With Thomas Strohmer.
- 2023 Fall
  - MAT 22A, Linear Algebra. With Martin Luu.
  - MAT 168, Optimization. With Luis Rademacher.
- 2023 Spring
  - MAT 21B, Integral Calculus. With Elena Fuchs.
  - MAT 150C, Modern Algebra. With Daping Weng.
- 2023 Winter
  - MAT 21D, Vector Analysis. With Blake Temple.
  - MAT 111, History of Mathematics. With William Tavernetti.

## Selected graduate coursework

**Algebra and topology.** Algebra (MAT 250A–C), Topology (MAT 215A–B), Differential Topology (MAT 239), Geometric Topology (MAT 216), and Lie Groups and Lie Algebras (MAT 261).

**Analysis and theoretical computer science.** Analysis (MAT 201A–B), Information Theory and Coding (EEC 266), and Computer Science Theory (ECS 289A).